Abirami J

AI Intern Candidate | Industrial AI, Machine Learning and Edge Deployment
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Summary

Final-year B.E. ECE student (2027) with a Minor in Artificial Intelligence from IIT Ropar, building AI for industrial and sensing systems: machine fault and anomaly detection, end-to-end Python ML pipelines, model optimization (INT8 quantization) and real-time deployment on embedded hardware. Research experience in deep learning and foundation models at IIT Madras.

Technical Skills

- **AI / Machine Learning:** Machine Learning, Deep Learning, Anomaly Detection, Fault Classification, CNN, Autoencoders, Random Forest, Transformers (ViT), Foundation Models, Model Evaluation, Ablation Studies, Explainable AI (SHAP)
- **Frameworks and Libraries:** PyTorch, TensorFlow, Keras, TensorFlow Lite Micro, scikit-learn, OpenCV, YOLOv8, timm, boto3
- **Programming:** Python, C, Embedded C, MATLAB
- **Deployment and Industrial IoT:** Edge AI, TinyML, INT8 Quantization, Real-Time Inference, Sensor Data Processing, ESP32-S3, Nordic nRF, BLE, UART, I2C, SPI
- **Signal Processing:** Feature Engineering, DSP Filtering, Time- and Frequency-Domain Features, MFCC, ECG Processing
- **Tools and Certifications:** Git, AWS S3, Bash, SSH, JupyterLab, Google Colab, KiCad; Microsoft Azure IoT Edge (2026); OpenCV Bootcamp (93%)

Industrial AI Projects

VAAZHVU: Edge AI Acoustic Fault Detection for Industrial Machines | Python, TensorFlow Lite, ESP32-S3
- Built an end-to-end machine condition-monitoring system: extraction of 70 time-, frequency- and MFCC-based features per 0.5 s frame, model training, INT8 quantization and real-time on-device inference.
- Combined an autoencoder anomaly detector with a supervised fault classifier in a dual-branch model, reaching 85.3% accuracy at 0 dB SNR (heavy background noise) with under 6% false alarms.
- Optimized the model to a 32 KB footprint and reduced wireless (BLE) data transfer by 91.2%.
- Presented at ICIDTSM 2026 (International Conference on Industrial Digital Twins for Smart Manufacturing), IIT Madras.

ArrhythmiaGuard: Real-Time Explainable Classification on Edge | Python, TensorFlow Lite, scikit-learn, ESP32-S3
- Trained and benchmarked a 1D CNN (AUC 0.908) and Random Forest (AUC 0.942) on sensor time-series under a strict inter-patient evaluation protocol (AAMI EC57), with SHAP explanations of each prediction.
- Debugged an undocumented TensorFlow Lite 2.19 zero-output failure in deployment and fixed it with hybrid INT8/float32 I/O. Presented at IEEE-WIE TechConnect 2026; Top 50 Teams, TNWise 2026.

Experience

Data Science Intern, Imaging Team Jun 2026 – Jul 2026
Sudha Gopalakrishnan Brain Centre, IIT Madras Chennai, India
- Built a Python framework fusing embeddings from five pretrained foundation models (GigaPath, Virchow, UNI2, PathDino, CONCH) with DINO teacher-student distillation for 11-class image classification (80.81% accuracy), with ablation studies.
- Built a memory-efficient pipeline streaming 1,557 image patches from AWS S3 (boto3); normalization cut colour variation by 40%.
- Delivered a classifier plugin for an annotation platform, trained on only 5 labelled slides, cutting manual effort by 60%.

R&D Engineer Intern Jan 2026 – Apr 2026
8bix Innovation Center, AIC Anna University Incubator Chennai, India
- Developed BLE firmware in Embedded C (Nordic nRF, GATT) with a UML-designed charge/therapy/fault/idle state machine for a wearable device; resolved field pairing and RSSI instability; designed power, protection circuits and BOM.

Electronics Test Intern Jun 2025 – Jul 2025
ETDC Chennai, STQC, Ministry of Electronics and IT, Government of India Chennai, India
- Tested and calibrated electronic and IoT devices against BIS, IEC and MIL-STD standards in a NABL-accredited laboratory (ISO/IEC 17025), including PCB-level fault identification.

Education

B.E. Electronics and Communication Engineering | CGPA 8.83/10 2023 – 2027
Meenakshi Sundararajan Engineering College (Anna University) | Minor in AI, IIT Ropar | Honours in Semiconductor Systems Chennai

Achievements

- Abstract accepted and presented, ICIDTSM 2026, IIT Madras: ge.iitm.ac.in/icidtsm-2026/abstract-book.pdf
- Provisional Patent Filed: AI-Powered Infant Cradle System (Indian Patent Office)
- First Prize, Paper Presentation, Crescent College, Chennai, 2026 | Academic Topper of the Year, 2023–24